

ADVANCED JAVA Syllabus To Be Covered During Second Phase of CRT
(18th to 28th May 2026)

Introduction:

- Core JAVA Vs Advanced JAVA
- Introduction to the components of advanced JAVA

JDBC (Java Database Connectivity)

- Introduction
- Connecting to Databases (MySQL)
- CRUD Operations using JDBC
- Types of Statements
- Transactions in JDBC

Java Servlets

- Introduction
- Creating Servlet using NetBeans IDE
- Working
- Create Servlet in MyEclipse IDE
- Life Cycle
- Request, Response, and Session
- Session Management
- Filters
- CRUD Operations
- Mini Project: Web Application Using Servlet
- Servlet Complete Tutorial

Hibernate (ORM Framework)

Hibernate simplifies database operations using Object-Relational Mapping (ORM) to map Java objects to database tables. Learn configuration, entity mapping, HQL, Criteria API and caching.

- Introduction
- Hibernate Setup & Configuration
- CRUD Operations
- Mapping (One-to-One, One-to-Many, Many-to-Many)
- Hibernate Annotations
- Hibernate Query Language (HQL)
- Hibernate Native SQL
- Caching
- Inheritance Mapping
- Mini Project: Hibernate using JPA and MySQL
- Hibernate Complete Tutorial

Spring Boot

- Introduction
- Getting Started with Spring Boot Application
- Architecture
- Spring Boot Annotations
- Spring Boot Configuration (Properties & YAML)
- Spring Boot with REST API

- RESTful Web Services
- REST Controller
- Create a REST API
- Testing REST APIs (with Postman or curl)
- Dependency Management
- Starters
- Exception Handling
- Actuator